Computer Architecture Lab #2

Objectives

After this lab, the student should:

Understand VHDL basics:

Multiple Architectures for same entity

Generic Entities

Use of (For … Generate)

Understand (When … Else) VS (With …Select)

Requirements

Design a n-bit ALSU that accepts two n-bit input values A and B and provides output F, the ALSU has 4 selection inputs S3, S2, S1, S0 and Cin input. The ALSU provides a total of 20 operations specified in the following table

	S_3	S_2	S_1	S_0	Cin =0	Cin = 1
Part A	0	0	0	0	F = A	F = A + 1
	0	0	0	1	F = A + B	F = A + B + 1
	0	0	1	0	F = A – B – 1	F = A – B
	0	0	1	1	F= A – 1	F = 0
Part B	0	1	0	0	Done Last Lab	
	0	1	0	1		
	0	1	1	0		
	0	1	1	1		
Part C	1	0	0	0		
	1	0	0	1		
	1	0	1	0		
	1	0	1	1		
Part D	1	1	0	0		
	1	1	0	1		
	1	1	1	0		
	1	1	1	1		

Deliverables :
Write VHDL code for part A in a separate VHDL files (don't forget to output the carry out)
You should use the full-adder given in the explanation instead of the VHDL operators (+) and (-)
Compile, your Code should be free of errors and warnings.
Simulate partA using Do file (**TestCases at the end of the Document**).
Bonus : optimized design (hint: you can use one full adder for part A).
Assignment: Integrate the 4 parts in one file And Make all of them **Generic and Simulate them using Do file**.

N.B. you will be graded for code neatness and understanding, Good luck

To test part A use the following table, let n = 8

Operation	A	B	Cin	F	Cout
F = A	0F	-	0	0F	0
F = A + B	0F	0001	0	10	0
	FF	0001	0	00	1
F = A – B – 1	FF	0001	0	FD	1
F= A – 1	FF	-	0	FE	1
F = A + 1	0E	-	1	0F	0
F = A + B + 1	FF	0001	1	01	1
F = A – B	0F	0001	1	0E	1
F = 0	F0	-	1	0000	0

To Test the Assignment

Operation	A	B	Cin	F	Cout
F = A	0F	-	0	0F	0
F = A + B	0F	0001	0	10	0
F = A + B	FF	0001	0	00	1
F = A – B – 1	FF	0001	0	FD	1
F= A – 1	FF	-	0	FE	1
F = A + 1	0E	-	1	0F	0
F = A + B + 1	FF	0001	1	01	1
F = A – B	0F	0001	1	0E	1
F = 0	F0	-	1	00	0
Operation	**A**	**B**	Cin	**F**	Cout
AND	F5	AA	-	A0	
OR	F5	AA	-	FF	
NOR	F5	AA	-	00	
NOT	F5	-	-	0A	
F=Logic shift right A	F5	_	-	7A	1
F=Rotate right A	F5	_	-	FA	1
F=Rotate right A with Carry	F5		0	7A	1
F=Rotate right A with Carry	F5		1	FA	1
F=Arithmetic shift right A	F5		_	FA	1
F=Logic shift left A	F5		_	EA	1
F=Rotate left A	F5		_	EB	1
F=Rotate left A with Carry	F5		0	EA	1
F=Rotate left A with Carry	F5		1	EB	1
F=0000	F5		_	00	0
F=Rotate right A	7A		_	3D	0